TRAINER KEY. DO NOT DISTRIBUTE. Day 3, Session 1: Worth the money? (economic model)

Source: H7-A. Base case: annual saving 812 AED (provisional, from the city DiD study); cost 1,800 AED at year 0; no saving in year 1; savings years 2 to 6; discount rate 5%. Values today: 737, 702, 668, 637, 606 (about 3,350 in total); ratio 3,350 / 1,800 = 1.86; NPV 3,350 - 1,800 = 1,550. **Scenario results (Q4, group-specific):** Card 1 (savings fade 30% a year) 1.07; Card 2 (+600 AED at year 0) 1.40; Card 3 (8% rate) 1.67; Card 4 (savings years 2 to 4) 1.17; Card 5 (fade + hidden costs) 0.80, the only one below 1. Among Cards 1 to 4, Card 1 lowers the ratio most (-0.79), then Card 4 (-0.69), Card 2 (-0.47), Card 3 (-0.19). Extra checks in §4: 40% discount rate gives 0.66; a three-year delay (years 5 to 9) gives 1.61.

Underlined = claim the source supports. ~~Struck out~~ = claim the source does not support. Green boxes = model answers (accept equivalent wording).

Page 1 | H7-A: The economic model summary

GreenWaste is a fictional training case. Use this model summary as your source for today's questions.

§1 | Where does the benefit input come from?

The city difference-in-differences study compares annual waste-cost changes for 4,794 participant businesses and 5,206 comparison businesses without GreenWaste, at 12 months. The difference is a reduction of **812 AED per participant business**. A causal reading needs parallel trends and the other conditions for the comparison. This is a provisional estimate, not a guarantee. Other methods compare different groups in different ways.

Table I: Inputs to the model and where each comes from

Input	Value	Source
Annual saving per participant business	812 AED	Estimate from the city study (provisional)
One-off cost per participant business	1,800 AED	Assumption in this fictional case
Year when savings start	Year 2	Assumption
Number of years that savings last	5 years	Assumption
Discount rate	5% a year	Assumption

§2 | How are future amounts combined?

The model assumes that equipment delivery costs **1,800 AED per participant business now (year 0)**, that there is no saving in year 1, and that the same annual saving is made in each of **years 2 to 6**. Future savings are discounted at an assumed rate of 5% a year, so that every amount is shown at its value in year 0 (its value today). The city data do not show five years of savings. They show only the first 12 months. For year *t*, the value today of the saving is the annual saving divided by 1.05 raised to the power *t*. Table F and Table O are on page 2. The calculations use the exact inputs, and the figures shown are rounded.

§3 | Whose perspective, and which costs and benefits?

This model mixes two points of view. It compares a cost to the authority (equipment delivery) with savings to businesses. It is not a full appraisal of the authority's budget, and it is not a full social cost-benefit analysis.

Table P: Who bears the costs and benefits, and what is included?

Who and what	Included in the model?	Evidence we would need
Authority: equipment delivery	Yes, as an assumed cost at year 0	Amounts and coverage from the programme's records
Authority: staff and administration	Not known	What is already included, the amount and the timing
Business time and maintenance; wider effects	Not known	The point of view, the resource costs and any transfers between actors

Check that a cost is not already counted before you add it. Financial savings are not automatically net benefits to society. Charges, transfers and wider effects need separate treatment.

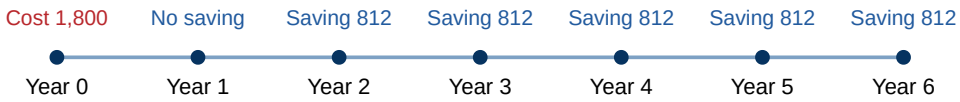
§4 | Sensitivity and the limits of the headline result

The scenarios change the model’s inputs. They do not change any observed fact about the programme. The settings for each scenario are on page 2. The scenario results table is given out after your group records its prediction. The results for Cards 1 to 4 depend on the ranges we tested, because each of those cards changes one input. Card 5 changes two inputs together. We have no evidence about how realistic these settings are.

An extra check shows that a discount rate of 40% would give a ratio of **0.66**, with all other inputs unchanged. Delaying the savings by three more years (years 5 to 9) at 5% gives a ratio of **1.61**, which is still above 1. These checks do not show which discount rate the government would use, or how long a delay is realistic. A ratio above 1, or a positive net present value, depends on the model and does not prove value for money. Our rule of at least a 1,000 AED reduction in annual waste management cost is a different test and is not a benefit-cost ratio threshold.

Source: synthetic GreenWaste city records and fictional economic assumptions. Inspect §§1 to 4, Tables I, P, F and O, and the scenario results table.

Page 2 | H7-A: Costs and savings by year, base case results and scenario settings



Assumed costs and savings by year: AED per participant business, before conversion to value today

Table F: Delivery cost and annual saving per participant business in each year, with each saving converted to its value today (AED)

Year	Cost (AED)	Saving (AED)	Saving at value today (AED)
0	1,800	0	0
1	0	0	0
2	0	812	737
3	0	812	702
4	0	812	668
5	0	812	637
6	0	812	606

One-year example: 812 divided by 1.05 is about 774 AED. The first saving in the model is in year 2: 812 divided by 1.05^2 is about **737 AED**. Add the values today for years 2 to 6. The rounded figures in the table may not add up exactly to the exact total.

Table O: Base case results per participant business: total benefits, cost, benefit-cost ratio and net present value (AED)

Measure	Value	What it is
Benefits at value today	3,350 AED	The savings in years 2 to 6, converted to year 0
Cost at value today	1,800 AED	The one-off equipment delivery cost at year 0
Benefit-cost ratio	1.86	Benefits divided by costs
Net present value	1,550 AED	Benefits minus costs

The benefit-cost ratio is benefits divided by costs. It has no units. The net present value is benefits minus costs, in AED per participant business. Neither number shows that GreenWaste caused the savings, that the savings will last, or that all costs are included.

Scenario settings: write your prediction before you see the results

- **Card 1: Savings fade.** After the first year of saving, each year's saving is 30% lower than the year before.
- **Card 2: Hidden costs.** We add 600 AED of costs at year 0 that are not already counted.
- **Card 3: A higher discount rate.** The rate is 8%.
- **Card 4: A shorter life.** Savings last three years (years 2 to 4).
- **Card 5: Savings fade and hidden costs.** This card changes the inputs of Cards 1 and 2 together.

On every card, all other inputs stay at the base values. Keep this page closed while your group makes its prediction. The scenario results table is given out after your group has recorded its prediction and its reason in Q4.

Page 3 | Q1 to Q3: inputs, timing and whose point of view

Q1 | What goes into the model? (§1 and §2, Table I)

a. Name one input that comes from the city study and one that is an assumption, and write where each appears.

> City study input: annual saving 812 AED (§1, Table I). Assumptions: e.g. 1,800 AED cost, the year savings start, five years, 5% discount rate (Table I; §2).

b. In which year does the cost occur? **Year 0 (now)**

c. In which year do savings start, and in which year do they end? Start: **Year 2** End: **Year 6**

d. How many years of savings are there? **5 years**

e. Is there a saving in year 1? **No (no saving in year 1)**

f. What evidence would you need before you trust the future savings?

> Evidence on whether savings continue beyond the first 12 months (the city data show only 12 months), e.g. later follow-up records, equipment lifetime and maintenance, whether the effect fades.

Q2 | Value today, ratio and net present value (Tables F and O)

a. Calculate the year 2 saving at its value today: 812 divided by **1.05² (= 1.1025)** is about **737** AED.

b. Why can we not simply multiply the annual saving by five?

> Savings start in year 2 and are received later, so each is worth less today. They are discounted by 1.05 raised to the year number; the sum of values today (3,350) is less than 5 x 812 = 4,060.

c. Write the benefits at value today: **3,350** AED. Write the cost at value today: **1,800** AED.

d. Show how the ratio is calculated: **3,350 / 1,800 = 1.86**

e. Show how the net present value is calculated: **3,350 - 1,800 = 1,550**

f. Explain what each of these two numbers means, under the model's assumptions.

> Ratio 1.86 (no units): under the model, benefits at value today are 1.86 times the cost. NPV 1,550 AED per participant business: benefits minus cost at value today. Neither shows that GreenWaste caused the savings, that they will last or that all costs are included.

Q3 | Whose point of view? (§3, Table P)

a. Whose decision is this?

> The authority (it pays for equipment delivery) or a decision-maker deciding whether to fund; the model mixes the authority's cost with savings to businesses, so the point of view must be stated.

b. Which costs and benefits belong to that point of view?

> From the authority's point of view: its costs (equipment delivery, staff and administration). Savings to businesses are a different point of view, and are not automatically a net benefit to society (transfers and wider effects need separate treatment).

c. Name one cost that may be missing.

> Examples: authority staff and administration; business time and maintenance; wider effects (Table P).

d. What might already be counted in the model?

> Equipment delivery at year 0 (1,800 AED) is counted. Staff and administration may already be in the 1,800 figure; that is not known.

e. What amount and timing would you request, without counting the cost twice?

> Request the programme's records of amounts, coverage and timing for each cost, and what the 1,800 AED already includes, so that costs are not double counted.

Page 4 | Q4 to Q6: scenarios and your recommendation

Q4 | Predict before you see the results

a. Write your card number and the input it changes. Card: **Group's own card (1 to 5), e.g. Card 3** Input: **e.g. discount rate (8% instead of 5%)**

b. Which inputs stay the same?

> Only the input named on the card changes (all other inputs stay at base values). On Card 5 two inputs change together.

c. Predict the benefit-cost ratio. Will it be above or below 1? Give your reason. After your group has recorded its prediction, you will receive the scenario results table.

Prediction: **Accept a reasoned prediction; compare with results: 1.07, 1.40, 1.67, 1.17, 0.80** Above or below 1? **Cards 1 to 4: above 1. Card 5: below 1**

> Accept a reasoned prediction. Card 3: the ratio falls modestly but remains above 1 because the savings are still larger than the cost. Cards 1, 2, 3 and 4: above 1; Card 5 falls below 1.

d. Write the ratio in the table: **Card 1: 1.07; 2: 1.40; 3: 1.67; 4: 1.17; 5: 0.80** . What explains the difference from your prediction?

> Explain gaps between the prediction and the table, e.g. for example Card 3 (8% rate) lowers the ratio only to 1.67 against the base 1.86, while Card 5 (fading savings plus hidden costs) falls to 0.80.

e. What evidence would show whether this change is realistic?

> Evidence on whether the change is realistic: e.g. Card 1: follow-up records on how savings last; Card 2: programme cost records for staff, administration and maintenance; Card 3: official discount-rate guidance; Card 4: equipment lifetime; Card 5: both.

f. Among Cards 1 to 4, which card lowers the ratio most? Why does this not give a ranking of the assumptions in general?

> Card 1 (ratio 1.07, a fall of 0.79). This compares only these settings (a 30% fade, 600 AED, 8%, 3 years); different ranges would change the order, so it is not a ranking of the assumptions in general.

Q5 | Read the model yourself (H7-A, pages 1 and 2)

a. Write one input that comes from the city study and one assumption, and where each appears.

> City study input: 812 AED saving (§1, Table I). Assumption: e.g. 1,800 AED cost, 5% rate, years 2 to 6 (Table I; §2). (Accept other valid pairs.)

b. What does the ratio support, under these assumptions?

> Under these assumptions, benefits at value today (3,350) exceed the cost (1,800): ratio 1.86 and NPV 1,550 AED per participant business, and the ratio stays above 1 in Cards 1 to 4.

c. What does it not show?

> It does not show that GreenWaste caused the savings, that savings last five years, that all costs are included, that it is value for money for society, or that the government would use a 5% rate. A ratio above 1 is not the same test as the 1,000 AED rule.

Q6 | Close the source and give a conditional recommendation

a. Explain what a benefit-cost ratio of 1.86 means.

> Under the model's assumptions, benefits at value today are 1.86 times the cost at year 0 for each participant business. It depends on a provisional saving, an assumed cost, assumed timing and an assumed rate.

b. State your provisional action and one condition.

> Example: "Ask first / act with conditions: I would continue only if later evidence supports lasting savings and the full costs are confirmed." Accept any defensible conditional action with a condition.

c. Request one specific missing fact, and explain how its answer could change your decision.

> Example: the full programme costs (staff, administration, maintenance). If they add 600 AED or more, the ratio falls to about 1.40 (Card 2), and combined with fading savings to below 1 (Card 5).

Page 5 | AI exercise: check the scenario

Stage 1: Mark the response

Naive prompt (a prompt with no context, as a busy person might type it): “Generate one plausible scenario where this programme’s benefit-cost ratio falls below 1. Tell me whether it is realistic.” Source: H7-A (the economic model summary). This response was written for the exercise. It is not a real AI answer.

> The base model gives a benefit-cost ratio of 1.86: discounted savings exceed the 1,800 AED cost. This is a useful starting point, although programme costs should still be checked.
>
> One scenario is to raise the discount rate to 40%. With all other inputs unchanged, the ratio is about 0.66, below break-even. ~~Such a rate is realistic and commonly used in government infrastructure appraisal, so this shows the programme should not be funded.~~
>
> ~~Alternatively, delaying benefits by a few more years also makes the ratio fall below 1.~~ ~~Both scenarios are plausible.~~ ~~The five-year saving stream in the source is established, so changing the discount rate is the main issue to investigate.~~

Keep the claims that H7-A supports. Cross out the arithmetic, timing or realism claims that it does not support. Write one source reference that changes the advice.

> Supported: the base ratio 1.86 (discounted savings exceed 1,800 AED), programme costs should still be checked, and a 40% rate gives about 0.66. Crossed out: that 40% is realistic and commonly used in government appraisal; that the programme should not be funded; that a few more years of delay brings the ratio below 1 (§4: 1.61); that both scenarios are plausible; that the five-year stream is established (it is an assumption); that the discount rate is the main issue. Source reference that changes the advice: §4 (a three-year delay gives 1.61, above 1) or Table I (five years is an assumption).

Stage 2: Run the improved prompt in Microsoft Copilot

Improved prompt

I am reviewing the fictional GreenWaste cost-benefit model H7-A below. Identify which inputs are estimated from data, modelled or assumed. Do not call the 1,800 AED cost measured, and do not call the difference-in-differences saving a proven causal benefit. Compare the four scenarios that each change one input (Cards 1 to 4). Say how far each one moves the ratio from the base ratio, and rank them only within these stated changes. Do not rank assumptions in general without defining their ranges. Card 5 changes two inputs together, so keep it separate. Before you recommend which assumption to investigate, ask me questions, including what evidence would make each change realistic and which discount-rate guidance applies. Keep the timing, the point of view, and the difference between the ratio and the net present value explicit. Cite the paragraph or table in H7-A. Here is the source: [paste worksheet pages 1 and 2 and the scenario results table, which you receive after your group has recorded its prediction].

Use only the information in H7-A, and keep your AI good-practice card beside you. If you are not using Copilot, read the prepared response on page 6 after you finish Stage 1.

a. Write one number you verified.

> Accept any verified number, e.g. ratio 1.86, NPV 1,550 AED, 737 AED for year 2, or 0.66 at 40%.

b. Write one useful question that the response asked about missing evidence.

> Example: "Which discount-rate guidance applies?" or "What evidence supports five years of savings?" Accept any question about missing evidence.

Page 6 | Prepared response

Use this response if you are not using Copilot. Read it after you have marked the response on page 5. It was written for the exercise. It is not a real AI answer. An AI tool may answer differently.

Separate inputs from conclusions

H7-A §1 and Table I use an **estimate from the city study** of about 812 AED, provisionally treated as a causal benefit. The cost, the one-year lag, the five-year stream of savings and the 5% rate are **assumptions**, not measured economic facts. Future benefits are modelled from them.

Tables F and O put costs at year 0 and savings in years 2 to 6. The benefits at value today are about **3,350 AED**, the cost is **1,800 AED**, the ratio is **1.86** and the net present value is **1,550 AED per participant business**. These numbers do not verify that GreenWaste caused the savings, how long they last, or that all costs are covered. Table P has a limited boundary that mixes two points of view. It is not a complete appraisal of the authority's budget or of society.

Quantify the stated changes

The scenario results table gives ratios of **1.07, 1.40, 1.67 and 1.17** for Cards 1 to 4. Compared with the base ratio of 1.86, the largest fall among those settings is the fading savings scenario (about **0.79** ratio units), then the shorter life (**0.69**), the added costs (**0.47**) and the rate of 8% (**0.19**). The ranking depends on the stated changes. Card 5 combines fading savings and added costs and gives **0.80**. It is not a test of one input.

At 40% the arithmetic can produce a ratio below 1 (§4). The model summary does **not** show that this rate is used in government practice. A delay of three more years gives a ratio of about **1.61**, which is not below 1. A stress test with a high rate does not show that the programme should be rejected. Suitable guidance and other evidence are needed.

Questions before recommending an input to investigate

- Whose decision is this, and which point of view should the appraisal take?
- What evidence on lasting results and maintenance supports five years of savings or a 30% fall each year?
- What costs, timing and existing coverage support the extra 600 AED?
- Which discount-rate guidance applies, and how credible is the provisional difference-in-differences benefit?

Verify this response yourself

a. Check one number against a named table.

> Example: ratios 1.07, 1.40, 1.67, 1.17 for Cards 1 to 4 (Scenario results table); base 1.86. Accept any number checked against a named table.

b. Write one request for evidence that could change the spending decision, and say why.

> Example: evidence on whether savings last or fade, because Card 1 shows the largest fall; or full cost records, because Card 5 falls below 1.